The first think that you need to accept if you want to be a trader is that learning programming is becoming unavoidable. Simply put, if you don't understand programming, you will be quite helpless in this game. If you think that programming is for programmers, that such menial tasks are beneath you, then the best of luck to you. If you believe that you can simply hire a programmer and tell him or her what to do, then you really have missed what this is all about. Either take this seriously, or stay away from it. Learning coding is unavoidable.

The good news is that you don't need to be an expert programmer. You need to know enough, and know how to figure stuff out when required. Knowing enough means that you need understand how backtesting works, the limitations of your software and methodology, and naturally a solid understanding of statistical analysis. If you understand these things and if you can code the strategy yourself, you will have a deep understanding of what your strategies do, of the underlying assumptions, the limitations, and the inevitable problems involved. This is the only way you can trust the rules, and if you can't trust the rules, you certainly won't be able to follow them when things get tough.

So you do need to figure out what kind of environment you need for your research and backtesting. A good environment for developing trading strategies should have an extreme level of flexibility, allowing you to do just about anything. Even things that the developers of that environment never imagined. It also needs to be able to handle large amounts of data and process such data quickly. We're crunching huge amounts of numbers here and most of us aren't ready to wait for hours for a backtest to be completed.

These requirements lead us down one single path. We need a sophisticated, powerful quant developing environment, one which is easy to learn, quick to develop in, and can handle fast analysis of large time series. While there are a few valid choices, there are plenty of more that just lead to a waste of time. Be sure to pick a proper, industry-standard platform, something which has considerable development support and isn't likely to simply go away any time soon. If you have no set choice, I would very much encourage you to go down the Python route.

Python and Zipline

From a finance and trading point of view, Python as a programming language is objectively special. It's not just yet another language with some